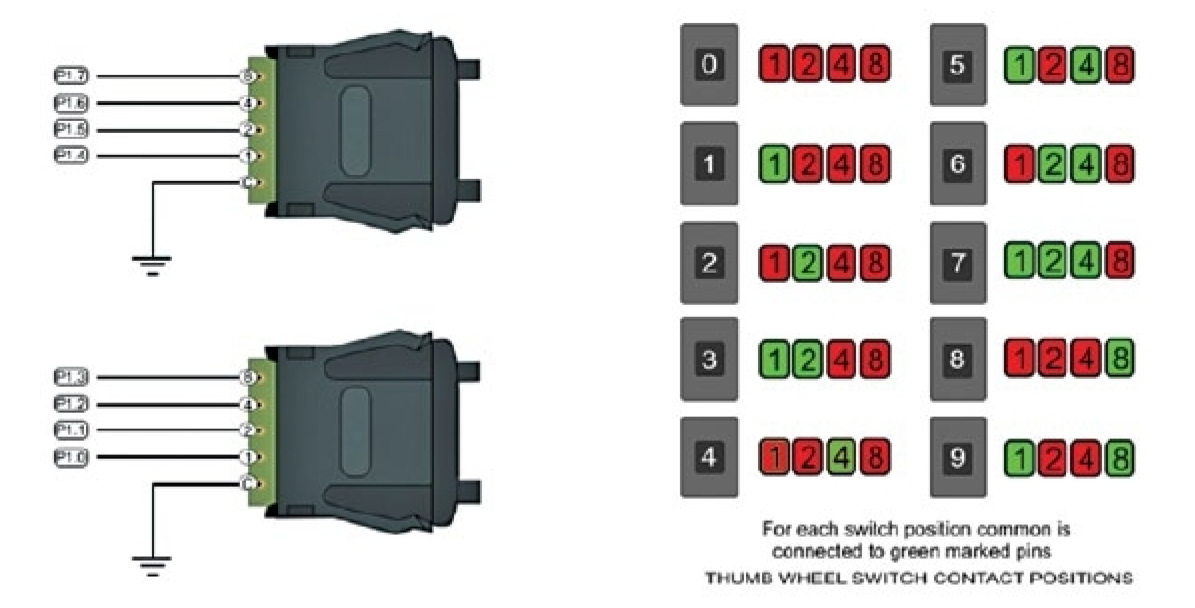

(a) Thumbwheel connections with MCU (b) Thumbwheel button positions

Fig. 2: Description of thumbwheel switches

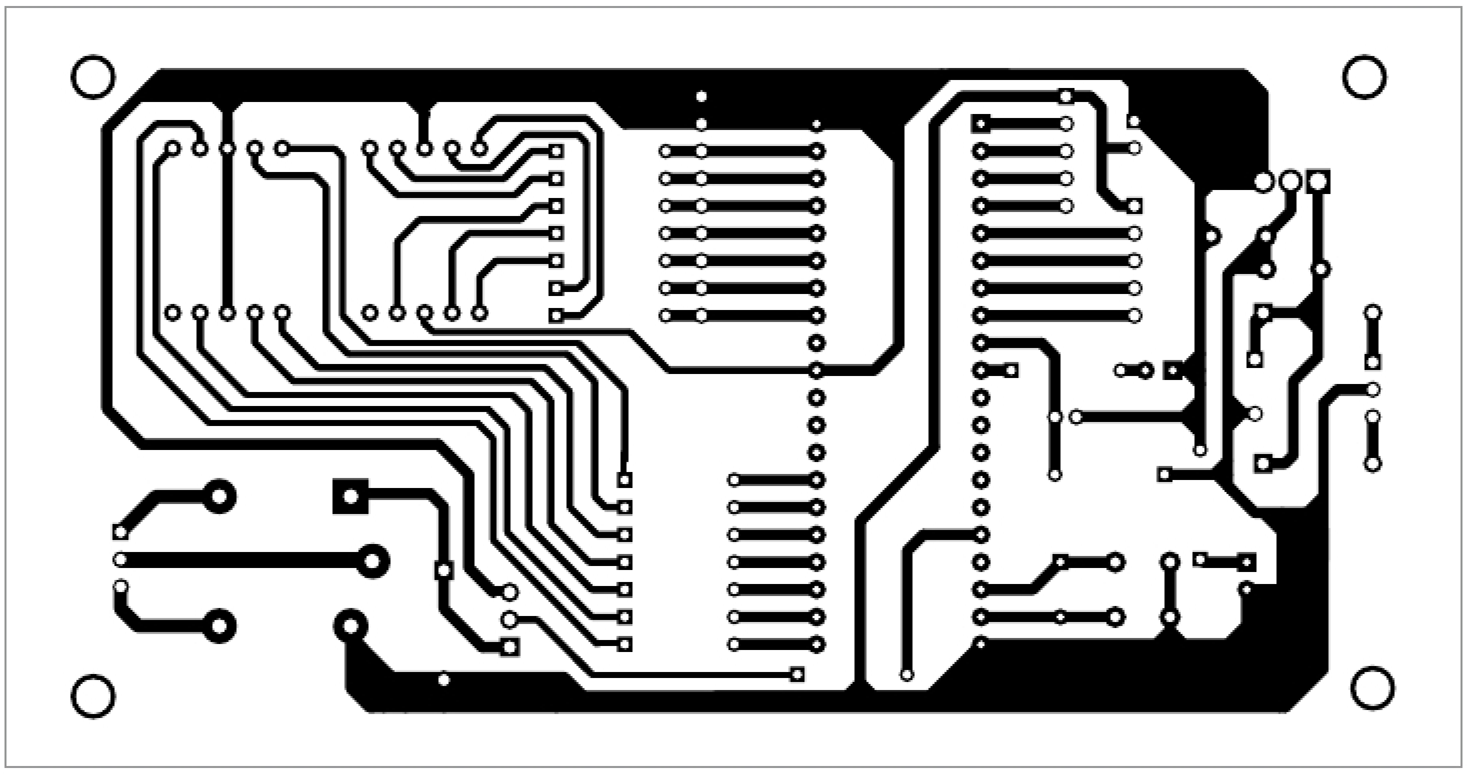

Fig. 3: Actual-size PCB layout of industrial timer

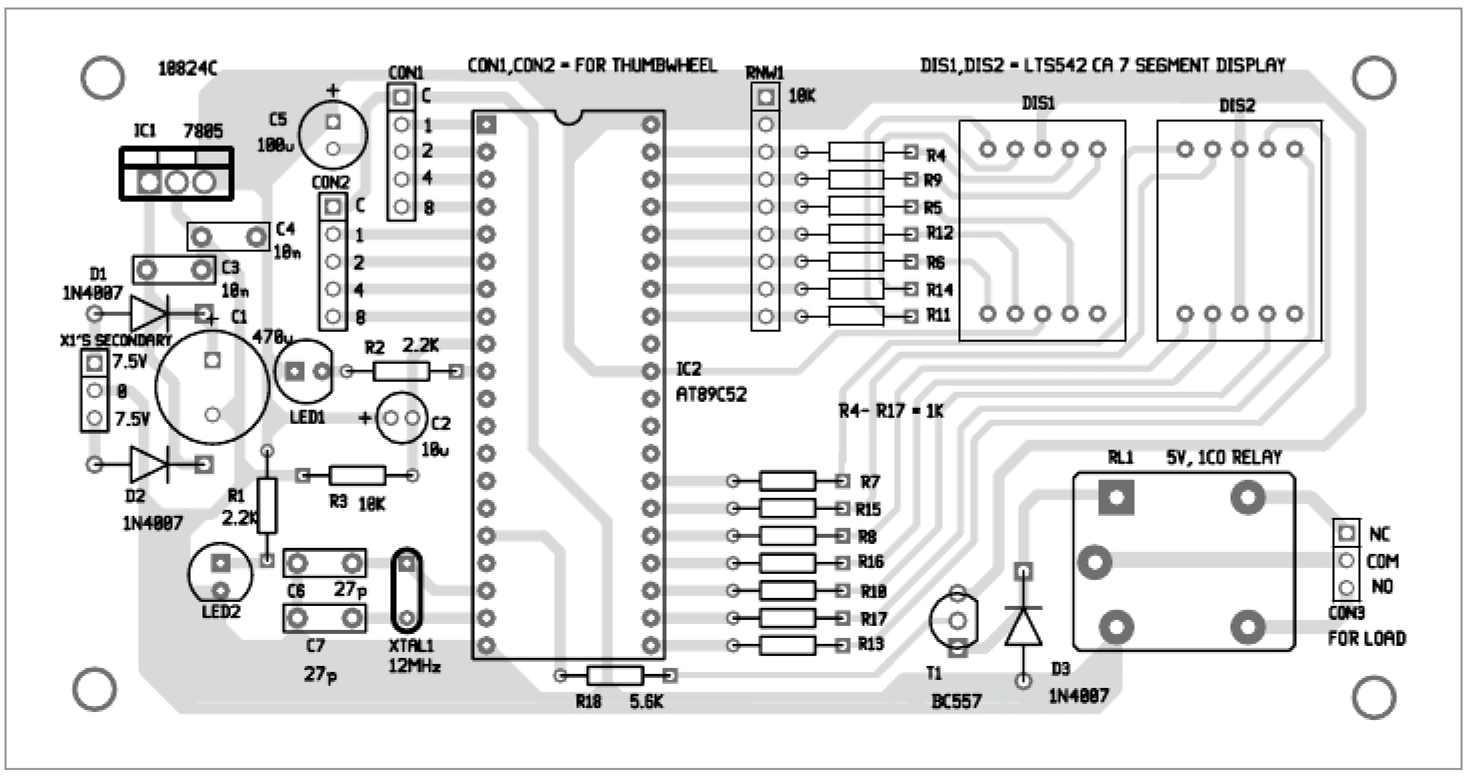

Fig. 4: Components layout for the PCB

to set the time as 13 seconds, first set 3 using TWS1 and then 1 using TWS2. Then, when you close S1, counter starts counting 01, 02, 03… till it reaches 13 and LED1 starts blinking. After displaying 13, the counting automatically stops, LED1 stops blinking and RL1 energises.

Software

The program code is written in C language and compiled with Kiel µvision 4 software. The main part of the program scans the settings of the thumbwheel switches and saves the data. The program starts incrementing count value from 0 to set value with a delay of one second and displays current values on the 7-segment display. RL1 gets activated on completion of the count and stays in that condition indefinitely as long as power is available.

The hex code generated using Keil software is burnt into the MCU using any suitable AT89C52 programmer. (TopView programmer was used by EFY Lab during testing.)

Construction and testing

An actual-size PCB layout for the industrial timer is shown in Fig. 3 and its components layout in Fig. 4.

Place the components on the PCB and solder these carefully. Load the hex file into the MCU and place it on the 40-pin IC base of the PCB. Connect the thumbwheel switches to the MCU as per circuit diagram.

After assembling the circuit, enclose it in a suitable box. Fix LED1, LED2, TWS1, TWS2, DIS1 and DIS2 on the front side of the